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5 February 1964
SUPERSEDING
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MILITARY SPECIFICATION

TRANSPARENCIES, ANIMATED; GENERAL SPECIFICATION FOR

This specification has been approved by the Office of Naval Research (U. S. Naval Training Device Center), Department of the Navy.

1. SCOPE

1.1 This specification covers animated transparencies for use with Overhead Projector, Master Vu-Graf, C. Beseler Company Catalog 3200 or equal, using a 1,000-watt bulb.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

PPP-B-621	Boxes, Wood, Nailed and Lock-corner
PPP-B-636	Box, Fiberboard

Military

MIL-P-116	Preservation; Methods of
MIL-L-10547	Liners, Case and Sheet, Overwrap; Water-vaporproof or Waterproof, Flexible

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

/FSC-6910/

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2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

THE SOCIETY OF THE PLASTICS INDUSTRY, INCORPORATED

Plastics, Engineering Handbook

(Copies may be purchased from Reinhold Publishing Corporation, New York.)

ROHM AND HAAS COMPANY

Manual No. PL-28L, Design and Fabrication Data, October 1962 edition

(Copies may be obtained upon application to Rohm and Haas Company, Philadelphia 5, Pennsylvania.)

3. REQUIREMENTS

3.1 Materials.- Unless otherwise specified herein, materials shall be as specified on figures 1 and 2. Materials not specified herein or on the figures shall be selected on the basis of compatibility with the established design and suitability to the specific application in satisfying the performance requirements specified herein.

3.1.1 All plastic used shall be acrylic or equivalent material.

3.1.1.1 Cements used for joining plastic surfaces shall be those recommended by the manufacturer of the plastic. Cementing shall be limited to the surfaces so treated in the Government-loaned laboratory model (see 3.10).

3.1.1.2 Lacquers or inks shall be those which will form a permanent bond with the material surface.

3.1.2 Except for springs, all metal parts and fastenings used in the device shall be of stainless steel, or be satin-finish chromium plated, or anodized clear, as applicable.

3.2 Design.- The animated transparency shall be capable of illustrating component arrangement and demonstrating operation of technologic principles and systems when used with the overhead projector. This shall be accomplished when light is projected through the baseplate and transparent components, and is masked by the opaque components. The components shall be positioned in such a manner as to simulate clearly the action of the components of the original equipment.

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3.2.1 Components.- The device shall consist of the following major components:

(a) Baseplate

(b) Movable components which, by their shape, color, action, and location, simulate parts and their action in mechanical or other assemblies

(c) Controls for actuating components; flaps for masking-off appropriate areas

(d) Carrying case.

3.3 Construction.-

3.3.1 The device shall be constructed in accordance with the Government-loaned laboratory model (see 3.10) and the requirements specified herein.

3.3.2 The laboratory development model is to be considered a sample. Construction methods or materials may be varied in order to permit more practical and economical manufacturing processes and tooling practices. The changes will be acceptable provided that (a) these changes do not adversely affect the color, optical clarity, training value or durability of the device and (b) the changes result in a part or assembly that can be disassembled or replaced by local fabrication in the event of loss or breakage. All changes shall be submitted in the form of preliminary drawings or data for written approval by the contracting officer prior to the construction of the device.

3.3.3 The transparency shall be constructed so that no fixed part or assembly becomes loose and no movable part, assembly or control becomes undesirably free or sluggish in operation. Mechanical performance shall not be impaired by shipment, operation or storage.

3.3.4 Cementing.- Cementing shall be used only in those cases where mechanical fastening is not practical. The cemented joints shall be free of air bubbles, and stress-crazing and shall have the transparency of the original plastic. The surfaces to be cemented shall be thoroughly softened by means of a solvent, so that a soft layer or cushion is produced on each, and the subsequent uniting and hardening will effect a strong, homogeneous bond. For recommendations on machining, finishing and cementing of acrylic plastics, publications listed in 2.2 may be used.

3.4 Performance.- The transparency, when used on the overhead projector, shall withstand the temperatures produced by a 1,000-watt bulb for a period of 45 minutes. At the end of this period, the transparency shall meet the following conditions:

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(a) Operate in accordance with 3.3.3
(b) No warpage which will cause seizing of moving parts
(c) No perceptible sag in any component when projected
(d) No visible discoloration or change in light transmission.

3.5 Details of components.-

3.5.1 Baseplate.- The baseplate shall be as specified on figure 1.

3.5.1.1 The white mask shall be integral with this baseplate and protected by a layer of clear plastic one-sixteenth of an inch thick.

3.5.1.2 The parting line between the layers of the baseplate shall be sealed so as to be dustproof.

3.5.1.3 The lower surface of the clear baseplate on which the components are assembled shall, at all times, be a minimum of one-sixteenth of an inch from the face of the projector lightplate to avoid abrasion of the plastic baseplate.

3.5.2 Movable components.-

3.5.2.1 All movable components shall be restrained and guided to prevent misalignment or breakage during normal classroom use. When packaged, they shall withstand shock and vibration incident to shipment by common carrier.

3.5.2.2 Overall height or thickness of the projectable components shall not exceed the height of the Government-furnished sample. The projected image shall be sharp and clear. Projectable movable components shall have a vertical separation to prevent abrasion of the moving parts.

3.5.2.3 Minimum thickness of any operable acrylic plastic component shall be one-sixteenth of an inch nominal.

3.5.2.4 Threaded portions of screws shall not be used as bearing surfaces for moving parts.

3.5.2.5 Screws shall be used for the assembly of moving parts in the same manner as in the laboratory model. They shall be secured to prevent loosening by rotation or movement of the parts. Rivets or pins shall not be used except as in the laboratory model.

3.5.2.6 Screws shall not extend beyond the thickness of the threaded material.

3.5.3 Focusing mark.- A projectable mark, in accordance with figure 1, shall be placed at the level and in a masked area away from the components where focus of this mark will give the optimum focus for the device.

3.5.4 Stops.- Flaps and slides shall be provided with stops which will prevent inadvertent overthrow or overtravel, and provide adequate support to prevent damage during operation and storage.

3.5.5 Electrical components.- Electrical components and wiring shall bear the label of or be listed by the Underwriters Laboratories (UL).

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3.5.6 Controls.-

3.5.6.1 Means of actuating the various components shall be as indicated in the laboratory model.

3.5.6.2 All controls shall be movable by normal finger pressure. No forcing shall be necessary during operation of the transparency.

3.5.7 Carrying case.- The carrying case shall be in accordance with figure 2 and as specified herein.

3.5.7.1 The carrying case shall have provisions to prevent movement of all loose and movable parts of the transparency. These provisions shall not cause any distortion or impair the operation of the transparency.

3.6 Dimensions and tolerances.- Dimensions shall be as shown on figures 1 and 2. Tolerances shall be such that the transparency will produce sharply defined projections and will satisfy the requirements of 3.4 and 3.5.3.2.

3.7 Finish and color.- Finish and color shall be as shown on figures 1 and 2 and as specified herein.

3.7.1 Finish.-

3.7.1.1 All transparent plastic components shall have polished upper and lower surfaces. All surfaces perpendicular to the baseplate shall have a clean machined or sanded finish equivalent to a 320 wet or dry finishing paper. These perpendicular surfaces shall not be polished or buffed.

3.7.1.2 All metal parts in contact with the movable plastic surfaces shall be smoothly finished to avoid abrasion of plastic parts.

3.7.1.3 Nonprojecting portions of the device, if required, shall be opaque and of uniform thickness and density.

3.7.2 Color.-

3.7.2.1 Colors for the components shall be the same as in the laboratory model.

3.7.2.2 Colors as shown in table I, or equivalents, shall be used.

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TABLE I - COLORS

COLOR	NUMBER
Green, light	2111
Green, medium	2082
Green, dark	2092
Blue, light	2069
Blue, medium	2083
Blue, dark	2045
Amber	2451
Yellow	2208
Red, light	2081
Red, dark	2444
Fluorescent, red	2085
Fluorescent, yellow	2086
Fluorescent, green	2154
Grey, light	2064
Grey, dark	2074
NOTE: Colors and their numbers are as listed by Rohm and Haas Company, Philadelphia.	

3.7.3 Cleaning.- All components of the transparency shall be thoroughly cleaned prior to assembly and shall be free of finger marks, smudges, or foreign particles of any kind.

3.7.4 Wax.- All surfaces of the transparency shall be waxed with an antistatic wax suitable for use with the type plastic used. The wax shall be polished dry and clear.

3.8 Nameplates or product markings.-

3.8.1 Nameplate.- A Government-furnished nameplate shall be affixed to the transparency in the same location as on the laboratory model.

3.8.2 The carrying case shall be marked in accordance with figure 2.

3.8.3 All nomenclature and notes shall be as on the laboratory model.

3.9 Government-furnished property.- A U. S. Naval Training Device Center nameplate will be furnished for each device.

3.10 Government-loaned property.- A laboratory model of the device will be loaned by the Government.

3.11 Workmanship.-

3.11.1 The device shall be constructed and finished in a thoroughly

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workmanlike manner. Particular attention shall be given to uniformity and neatness of all components to insure clean, sharp, projected images.

3.11.2 All projectable surfaces of all plastic components shall be free from scratches and abrasions.

3.11.3 All sharp corners and edges shall be removed from parts outside the projected area which may be handled during classroom use.

3.11.4 All metal parts shall be free from burrs and sharp edges. The threaded fasteners shall be cut to size and chased, prior to insertion in the part.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Inspection.- The inspection shall include such visual and mechanical examination, and testing of materials, subassemblies, parts and accessories (including purchased standard items) during the manufacturing process as may be required to assure conformance to all the requirements specified herein. The following inspections, consisting of the examinations and tests specified herein, shall be performed to determine compliance with the requirements of this specification:

(a) Preproduction inspection

(b) Production inspection.

4.2.1 Preproduction inspection.- Preproduction inspection shall be conducted on the preproduction sample (see 6.3.1) and shall consist of all the examinations and tests of 4.3.1.

4.2.2 Production inspection.- Production inspection shall be performed on production units to the extent necessary to maintain quality control following acceptance of the preproduction sample. The approved preproduction sample shall be used as a representative sample in the fabrication and testing of the production units to be submitted for acceptance.

4.3 Inspection procedures.- Inspection procedures shall be in accordance with the following:

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4.3.1 Examinations and tests.-

4.3.1.1 Examinations.- The animated transparency shall be examined for compliance with the following requirements as specified herein:

(a) Material

(b) Construction

(c) Performance

(d) Finish and color

(e) Nameplates and product markings

(f) Workmanship

(g) Preservation, packaging, packing and marking.

4.3.1.2 Tests.- The following operating test shall be conducted in order to determine whether or not the animated transparency conforms to the requirements of this specification.

4.3.1.2.1 Operating test.- Each animated transparency shall be subjected to an operating test on the Master Vu-Graph, C. Beseler Company Catalog 3200 or equal, using a 1,000-watt bulb for 45 minutes with heat filter removed. The animated transparency shall be capable of meeting the performance requirements of 3.4.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.- Preservation and packaging shall be level A or C (see 6.2).

5.1.1 Level A.- For level A, each device shall be preserved and packaged for mechanical and physical protection in accordance with Specification MIL-P-116, Method III.

5.1.1.1 Unit packaging.- Each device shall be packaged one unit to a container conforming to Specification PPP-B-636, Class 1.

5.1.2 Level C.- For level C, preservation and packaging shall conform to manufacturer's commercial practice.

5.2 Packing.- Packing shall be level A or C (see 6.2).

5.2.1 Level A.- Units packaged in accordance with 5.1.1 shall be packed in exterior containers conforming to Specification PPP-B-636 or PPP-B-621. Unless otherwise specified by the procuring activity (see 6.2),

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the gross weight shall not exceed the weight limitations of the shipping container. Each exterior shipping container shall be lined with a waterproof moisture barrier sealed with adhesive or tape to effect water resistance in accordance with Specification MIL-L-10547. A case liner shall not be required where the package(s) and package dunnage have been treated to resist water. Unit fiber boxes, conforming to Specification PPP-B-636, class 2, closed, reinforced and waterproofed as specified in the appendix thereto, and used as shipping containers, shall not be overpacked.

5.2.2 Level C.- Units packaged in accordance with 5.1.2 shall be packed in containers which will insure acceptance by common carrier and safe delivery at destination. Shipping containers shall comply with common carrier rules and regulations as applicable to the mode of transportation and may be the supplier's commercial practice when such meets the requirements of this level.

5.3 Physical protection.- All material shall be packaged and packed in containers constructed to provide a compact and nonshifting load. Adequate cushioning, blocking and bracing shall be provided within shipping containers in a manner designed to prevent damage due to shock, vibration or impact during shipment, handling and storage.

5.4 Marking.- Interior packages and exterior shipping containers shall be marked in accordance with Standard MIL-STD-129.

6. NOTES

6.1 Intended use.- The animated transparency training aids will be used by instructors in classroom training to demonstrate function and operation of principles and systems.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification

(b) Selection of applicable levels of preservation, packaging and packing (see 5.1 and 5.2)

(c) Gross weight of shipping container (see 5.2.1).

6.3 Definition.- Except as specified herein, definitions will be furnished upon written request to the procuring activity.

6.3.1 Preproduction sample.- A production item furnished under production contracts for the purpose of determining compliance with requirements of the detail training device specification already established and accepted. It shall be electrically and mechanically representative of production sets though not necessarily made with production line jigs, fixtures and methods.

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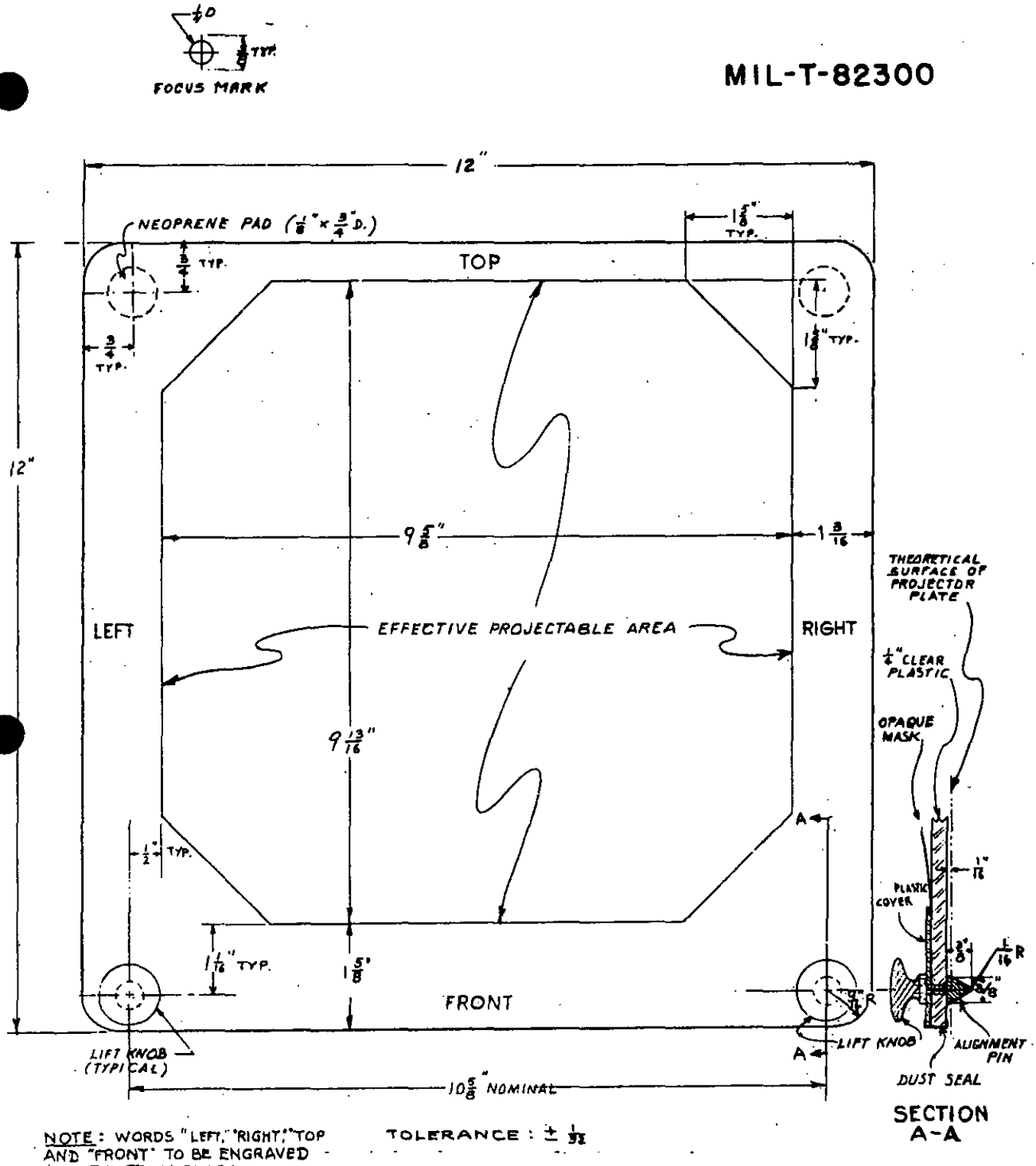
6.3.2 Production unit.- A unit in its final mechanical and electrical form of final production design made by production tools, jigs, fixtures and methods.

6.3.3 Training aid.- A training device designed to demonstrate or illustrate a concept.

6.4 Availability of Government-loaned laboratory model.- The laboratory model will be available for examination at the U. S. Naval Training Device Center (Code 3313), Port Washington, New York, and will be loaned to the contractor.

Preparing activity:
Navy-ONR(NTDC)

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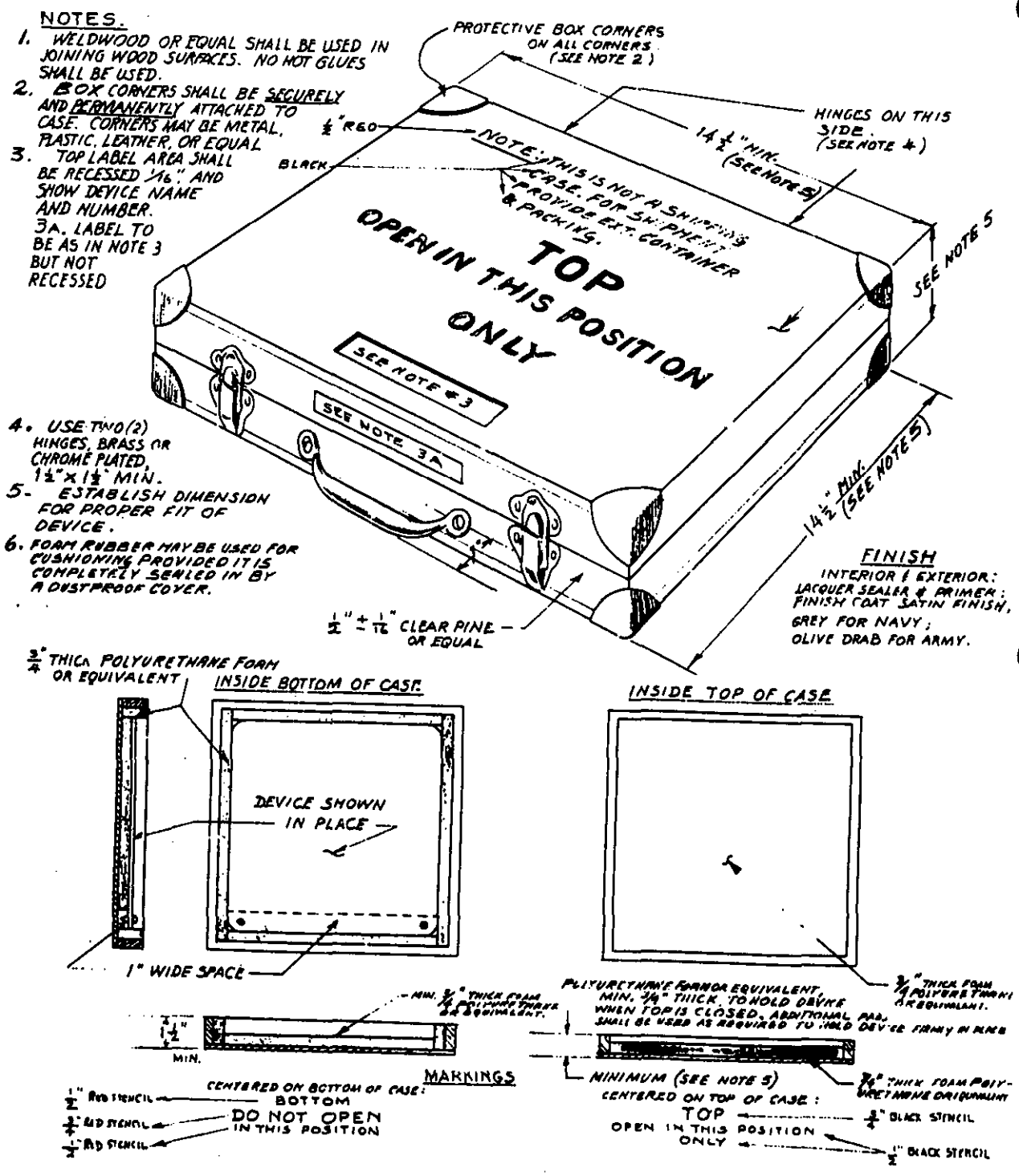
NOTE: WORDS "LEFT," "RIGHT," "TOP" AND "FRONT" TO BE ENGRAVED AND FILLED IN BLACK.

TOLERANCE: ± 1/32

TYPICAL BASEPLATE
FOR
ANIMATED TRANSPARENCY

Figure 1

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STORAGE/CARRYING CASE FOR ANIMATED TRANSPARENCY

Figure 2

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